

```
[5]: cat("Placelytics: A Data-Driven Analysis of Student Placement Patterns")
```

Placelytics: A Data-Driven Analysis of Student Placement Patterns

```
[7]: library(readr)
library(dplyr)
library(ggplot2)
```

```
[8]: data <- read.csv("dataset_aiplacement.csv")
```

```
head(data)
```

A data.frame: 6 × 8

	CGPA	Internships	Projects	Certifications	Aptitude	Chances	PlacementStatus	Improvement
	<dbl>	<int>	<int>	<int>	<int>	<chr>	<chr>	<chr>
1	9.09	2	3	2	91	High	Placed	Maintain consistency
2	9.25	2	4	2	92	High	Placed	Maintain consistency
3	9.23	2	4	3	77	High	Placed	Maintain consistency
4	8.83	3	4	3	84	High	Placed	Maintain consistency
5	8.45	3	3	2	87	High	Placed	Maintain consistency
6	9.25	2	4	2	96	High	Placed	Maintain consistency

```
[9]: str(data)
```

```
'data.frame':  10000 obs. of  8 variables:
 $ CGPA      : num  9.09 9.25 9.23 8.83 8.45 9.25 9.5 8.95 8.91 9.73 ...
 $ Internships : int  2 2 2 3 3 2 2 3 3 2 ...
 $ Projects   : int  3 4 4 4 3 4 4 3 4 3 ...
 $ Certifications : int  2 2 3 3 2 2 3 3 2 2 ...
 $ Aptitude    : int  91 92 77 84 87 96 98 97 80 82 ...
 $ Chances     : chr   "High" "High" "High" "High" ...
 $ PlacementStatus: chr   "Placed" "Placed" "Placed" "Placed" ...
 $ Improvement  : chr   "Maintain consistency" "Maintain consistency" "Maintain consistency" "Maintain consistency" ...
```

```
[10]: summary(data)
```

CGPA		Internships		Projects		Certifications	
Min.	: 5.000	Min.	:0.000	Min.	:1.000	Min.	:0.0000
1st Qu.:	5.890	1st Qu.:	0.000	1st Qu.:	1.000	1st Qu.:	0.0000
Median :	6.460	Median :	1.000	Median :	1.000	Median :	0.0000
Mean :	6.905	Mean :	1.199	Mean :	1.872	Mean :	0.8723
3rd Qu.:	8.100	3rd Qu.:	2.000	3rd Qu.:	3.000	3rd Qu.:	2.0000
Max.	:10.000	Max.	:3.000	Max.	:4.000	Max.	:3.0000

Aptitude		Chances		PlacementStatus		Improvement	
Min.	: 42.00	Length:	10000	Length:	10000	Length:	10000
1st Qu.:	61.00	Class :	character	Class :	character	Class :	character
Median :	71.00	Mode :	character	Mode :	character	Mode :	character
Mean :	73.22						
3rd Qu.:	87.00						
Max.	:100.00						

```
[11]: colnames(data)
```

```
'CGPA' · 'Internships' · 'Projects' · 'Certifications' · 'Aptitude' · 'Chances' · 'PlacementStatus' · 'Improvement'
```

```
[12]: colSums(is.na(data))
```

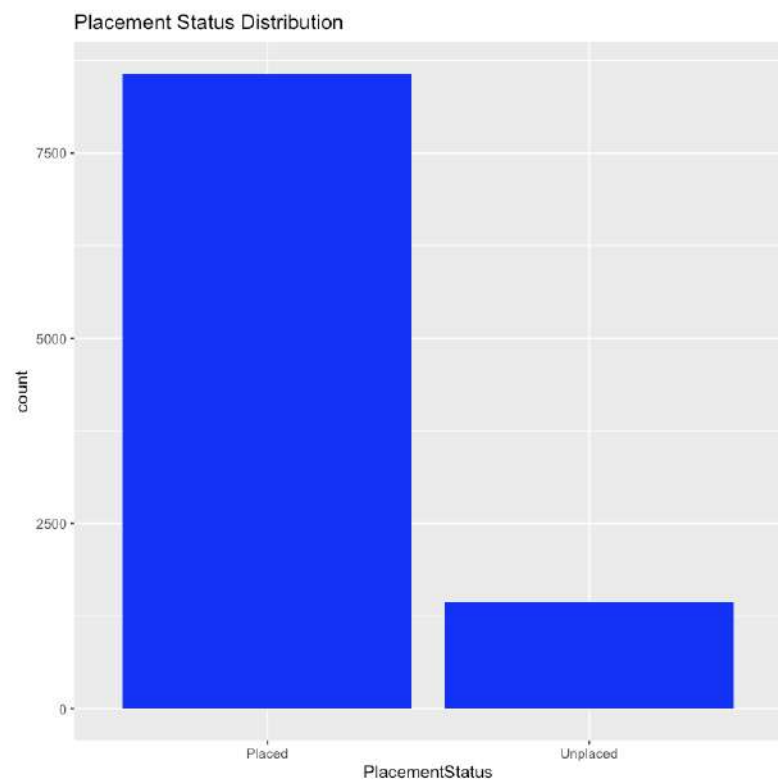
```
CGPA: 0 Internships: 0 Projects: 0 Certifications: 0 Aptitude: 0 Chances: 0 PlacementStatus: 0 Improvement: 0
```

```
[13]: data <- na.omit(data)
```

```
[14]: table(data$PlacementStatus)
```

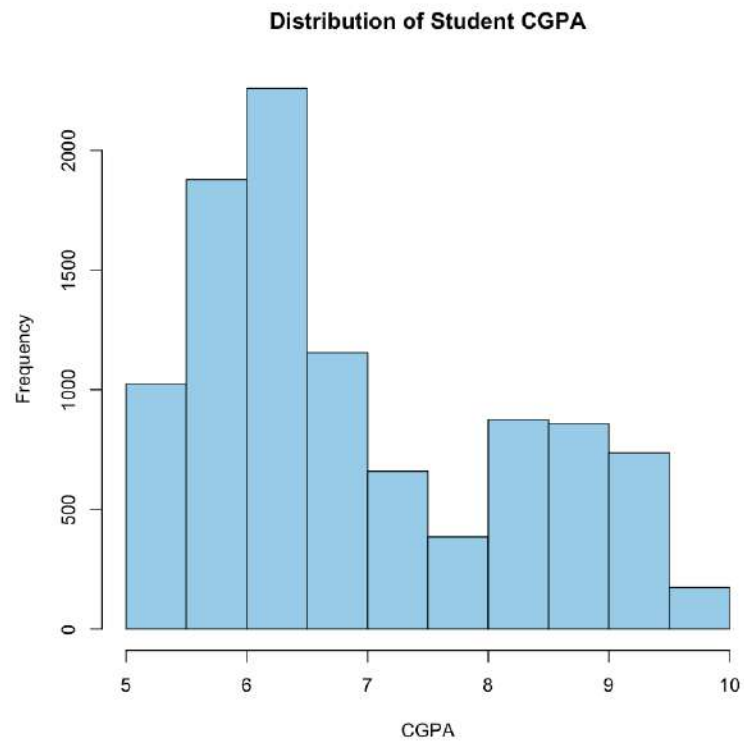
Placed	Unplaced
8570	1430

```
[15]: ggplot(data, aes(x = PlacementStatus)) +  
      geom_bar(fill = "blue") +  
      ggtitle("Placement Status Distribution")
```

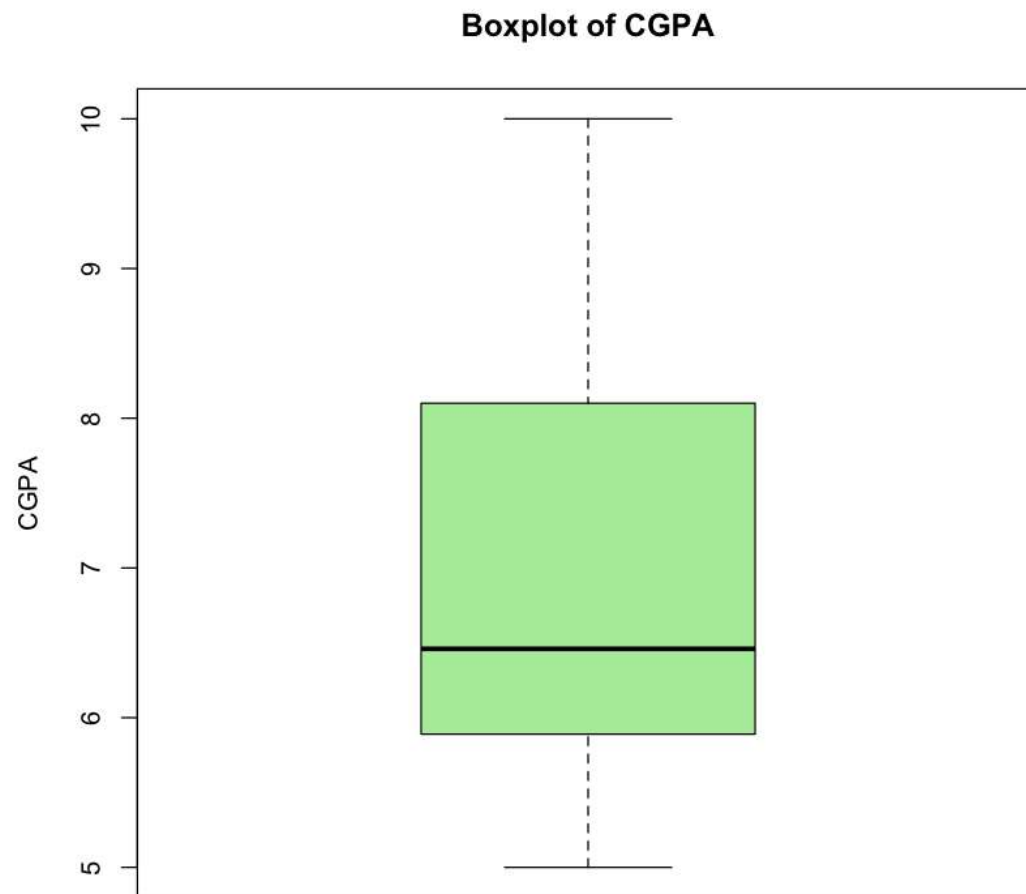


PlacementStatus

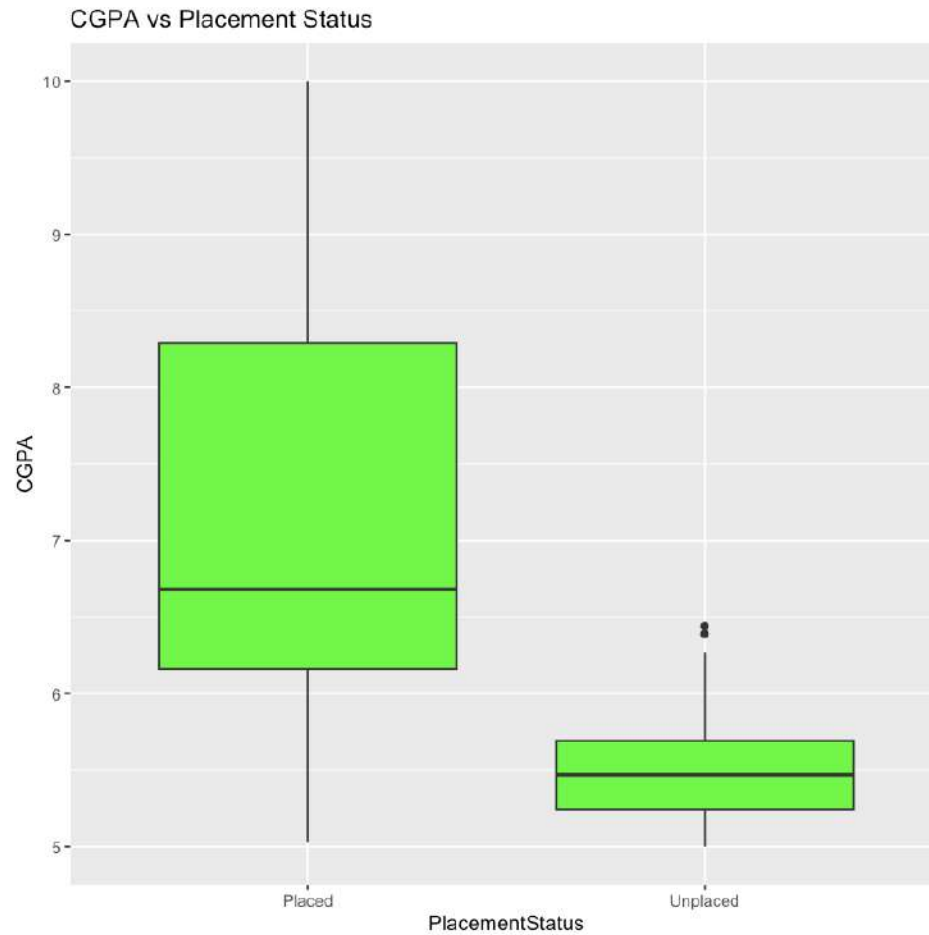
```
[16]: hist(data$CGPA,  
          col="skyblue",  
          main="Distribution of Student CGPA",  
          xlab="CGPA")
```



```
[17]: boxplot(data$CGPA,  
             col="lightgreen",  
             main="Boxplot of CGPA",  
             ylab="CGPA")
```



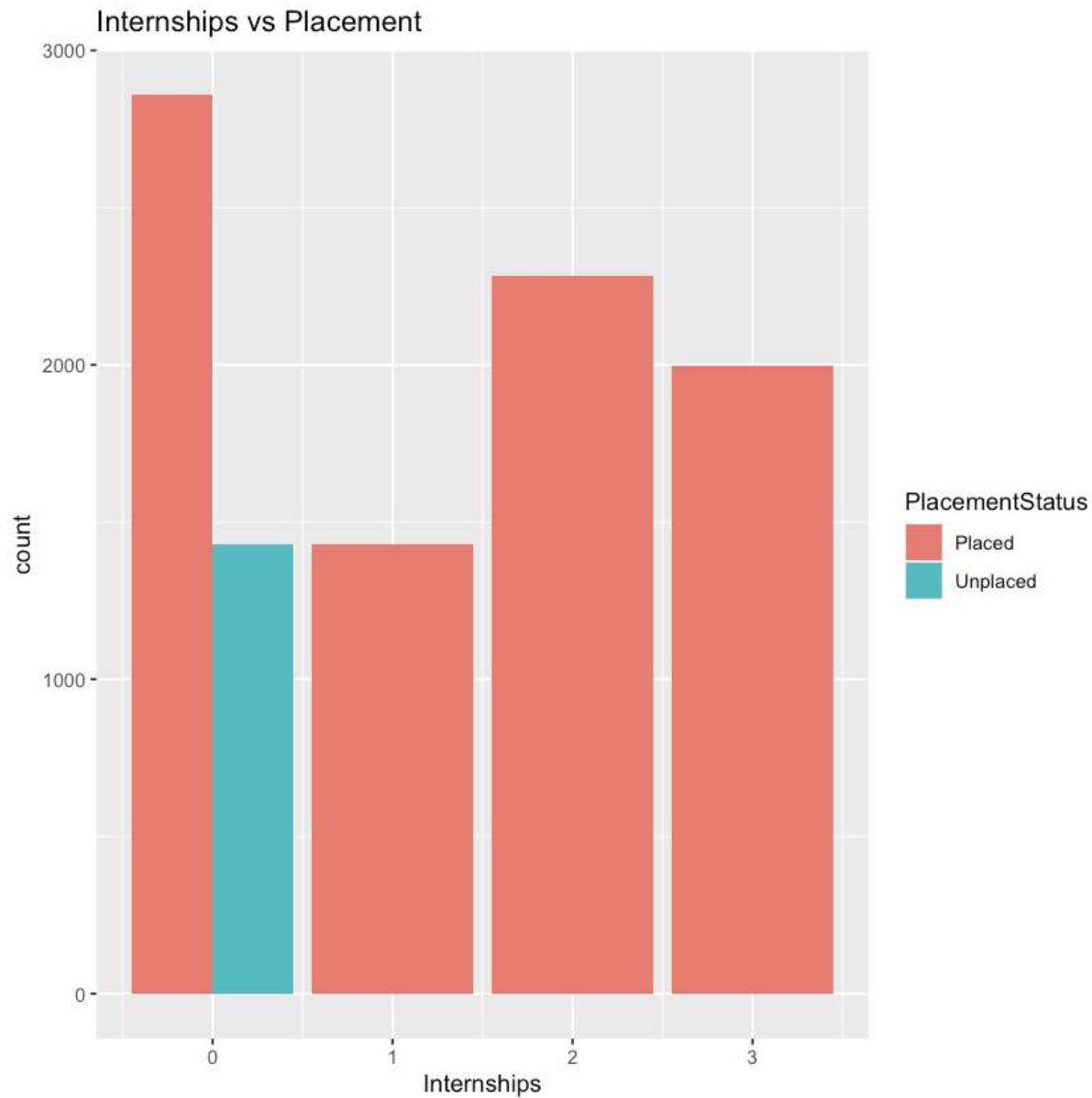
```
[18]: ggplot(data,
  aes(x = PlacementStatus,
      y = CGPA)) +
  geom_boxplot(fill="green") +
  ggtitle("CGPA vs Placement Status")
```



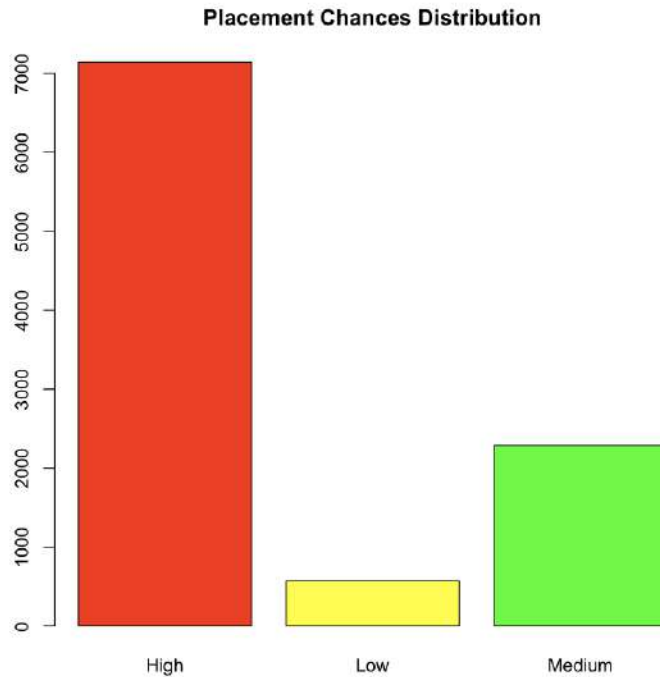
```
[19]: table(data$Internships,
  data$PlacementStatus)
```

	Placed	Unplaced
0	2860	1430
1	1430	0
2	2283	0
3	1997	0

```
[20]: ggplot(data,  
        aes(x = Internships,  
            fill = PlacementStatus)) +  
  geom_bar(position="dodge") +  
  ggtitle("Internships vs Placement")
```



```
[21]: barplot(table(data$Chances),
  col=c("red","yellow","green"),
  main="Placement Chances Distribution")
```



```
[22]: numeric_data <- data %>%
  select(CGPA,
    Internships,
    Projects,
    Certifications,
    Aptitude)

cor(numeric_data)
```

A matrix: 5 × 5 of type dbl

	CGPA	Internships	Projects	Certifications	Aptitude
CGPA	1.0000000	0.1421940	0.3252187	0.2659816	0.1410941
Internships	0.1421940	1.0000000	0.6384314	0.6489183	0.3163411
Projects	0.3252187	0.6384314	1.0000000	0.9564221	0.5152617
Certifications	0.2659816	0.6489183	0.9564221	1.0000000	0.5119438
Aptitude	0.1410941	0.3163411	0.5152617	0.5119438	1.0000000


```
[23]: colMeans(data[, c("CGPA",  
                        "Internships",  
                        "Projects",  
                        "Certifications",  
                        "Aptitude"])]
```

CGPA: 6.905029000000001 Internships: 1.1987 Projects: 1.8723 Certifications: 0.8723 Aptitude: 73.2234

```
[24]: model <- lm(Aptitude ~ CGPA,  
                data = data)  
  
summary(model)
```

Call:

```
lm(formula = Aptitude ~ CGPA, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-29.722	-11.987	-2.079	13.414	29.674

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	62.1388	0.7913	78.53	<2e-16 ***
CGPA	1.6053	0.1126	14.25	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

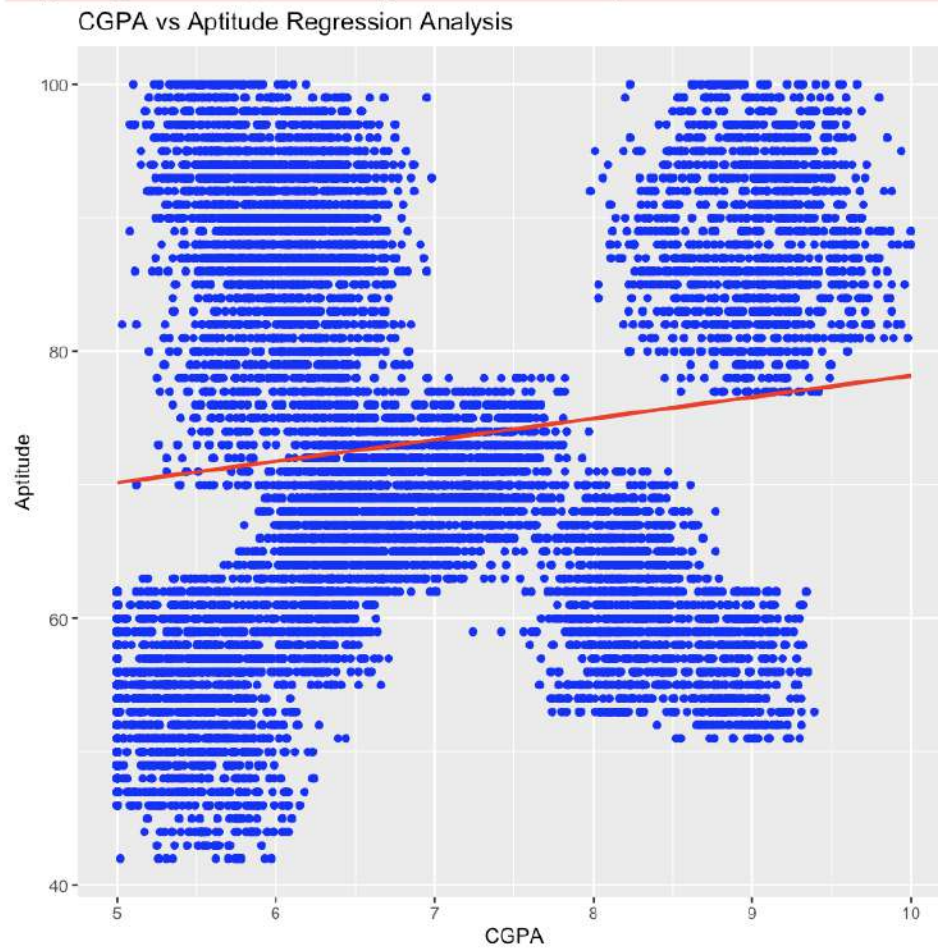
Residual standard error: 14.51 on 9998 degrees of freedom

Multiple R-squared: 0.01991, Adjusted R-squared: 0.01981

F-statistic: 203.1 on 1 and 9998 DF, p-value: < 2.2e-16

```
[29]: ggplot(data,
      aes(x = CGPA,
          y = Aptitude)) +
  geom_point(color = "blue") +
  geom_smooth(method = "lm",
              color = "red",
              se = FALSE) +
  ggtitle("CGPA vs Aptitude Regression Analysis")
```

`geom\_smooth()` using formula = 'y ~ x'



```
[30]: cor(data$CGPA,
      data$Aptitude)
```

0.141094077295259

```
[33]: predictPlacement <- function() {  
  
  cgpa <- as.numeric(readline("Enter CGPA: "))  
  internships <- as.numeric(readline("Enter Internships: "))  
  projects <- as.numeric(readline("Enter Projects: "))  
  certifications <- as.numeric(readline("Enter Certifications: "))  
  aptitude <- as.numeric(readline("Enter Aptitude Score: "))  
  
  score <- cgpa +  
    internships +  
    projects +  
    certifications +  
    aptitude  
  
  if(score >= 18){  
    print("Placement Status: Placed")  
    print("Chance: High")  
  }  
  
  else if(score >= 12){  
    print("Placement Status: Placed")  
    print("Chance: Medium")  
  }  
  
  else{  
    print("Placement Status: Unplaced")  
    print("Chance: Low")  
  }  
}
```

```
[34]: predictPlacement()
```

```
Enter CGPA: 8  
Enter Internships: 1  
Enter Projects: 2  
Enter Certifications: 3  
Enter Aptitude Score: 7  
[1] "Placement Status: Placed"  
[1] "Chance: High"
```